Thomas J. Barrett, PhD

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TECHNICAL SKILLS

Simulation & Modeling: Molecular Dynamics (GROMACS, LAMMPS), Density Functional Theory (ORCA), FEA (Abaqus, ANSYS, COMSOL), Molecular Docking (AutoDock Vina, gnina)

Machine Learning & Software: Python (PyTorch, E3NN, RDKit), C++, Fortran, Bash; GNNs, Random Forest, Diffusion/Flow-Matching Models

Experimental: AFM, WAXD, FTIR-ATR, UV-Vis, Raman, SEM, DMA, TGA, DSC, Tribometry, Nanoindentation, Electrospinning, Wet/Gel Spinning, Design of Experiments (DOE)

Additional: CAD, ONNX, 3Dmol.js, ESP32 (SoC), Docker, Git

EXPERIENCE

R&D Engineer Consultant · Applied Plastics | Norwood, MA — Feb. 2026 – Present

- Developed non-PTFE polymer-composite lubricious coatings for reel-to-reel wire coating processes, characterizing thin-film performance via **AFM, tribometry, Raman microscopy, and nanoindentation** to benchmark against incumbent PTFE products.
- Built a **Python-based inverse design framework** coupling surrogate models with Bayesian and evolutionary optimization to accelerate coating formulation; reduced experimental iteration cycles by targeting high-performance parameter space directly.
- Developed computational tribology models to normalize wear behavior across wire diameters using an Anton-Paar TRB3, enabling standardized comparisons across product lines.

Postdoctoral Researcher · Christian-Albrechts-Universität zu Kiel | Kiel, Germany — Feb. 2025 – Jan. 2026

Chair for Bioinspired Materials & Biosensor Technologies · Advisor: Prof. Dr. Zeynep Altintas

- **Designed and implemented a generalized polymerization algorithm** (Python/RDKit) wrapping GROMACS and AutoDock Vina to automate MIP biosensor screening, docking, polymerization, and assessment; natively supports **>100 monomers and 15,000+ reactions**, reducing manual recipe development from weeks to hours.
- Developed **Random Forest classifier and regression models** to predict MIP recipes for ion and small-molecule targets from DFT-derived molecular fingerprints; achieved comparable accuracy to full MD screening in **seconds vs. month-long** deterministic workflows.
- Formulated PCL/PLA and PCL/PLA/ZIF-8 polymer composite scaffolds for tricuspid valve replacement; optimized process to remain **stable and repeatable at 4× production scale**, validated through mechanical testing, cell proliferation studies, and FEA-guided shape optimization.
- **Co-authored a pending patent** (Germany) on computational design of imprinted polymer materials for small-molecule sensing.

R&D Intern · Applied Plastics | Norwood, MA — Oct. 2024 – Jan. 2025

- Applied **Design of Experiments (DOE)** to optimize PTFE reel-to-reel coatings; led mechanical characterization (scratch, CoF, wear, nanoindentation), resulting in a **new product line** that met performance targets while staying within existing FDA-approved device parameters, avoiding costly recertification.

Graduate Research Assistant · Northeastern University – MINUS Laboratory | Boston, MA — Jan. 2019 – Aug. 2024

Advisor: Dr. Marilyn Minus

- Built a **hierarchical multiscale modeling pipeline** (Python/C++/Fortran) converting LAMMPS MD microstructures into Abaqus-compatible meshes via crystallinity cluster analysis, enabling quantitative study of interphase regions in semicrystalline polymer–CNT composites.
- Performed end-to-end experimental characterization of cast films and gel-spun fibers using **WAXD, DMA, TGA, DSC, and SEM**, correlating microstructure to bulk mechanical properties.
- Published two first-author papers in *Journal of Chemical Theory and Computation* (2025): (1) Nose–Hoover barostat integration effects on uniaxial tension MD results; (2) *pothos*, an open-source Python package for polymer chain orientation and microstructure monitoring.

Graduate Research Assistant · Northeastern University – Applied (Bio)Mechanics & Tribology Lab Sep. 2016 – Dec. 2018

Advisor: Dr. Sinan Müftü

- Developed a **COMSOL multiphysics model** of sub-centimeter surface roughness effects on intertidal zone heat retention; demonstrated a strong correlation between RMS roughness and temperature extremes with implications for coastal ecology modeling.

EDUCATION

PhD, Mechanical Engineering (Materials Science) · Northeastern University 2019 – 2024

MS, Mechanical Engineering (Mechanics) · Northeastern University 2016 – 2018

BS, Mechanical Engineering · University of Rochester 2012 – 2016

SELECTED FIRST AUTHOR PUBLICATIONS & PATENTS

Barrett TJ, Moldovean-Cioroianu NS, Altintas Z. A digital twin for the in-silico screening, polymerization, and rebinding of Molecularly Imprinted Polymers. Awaiting submission.

Barrett TJ, Saeid Nia, M, Seiler, J-A, Sharifuzzaman, M, Altintas Z. Indications of ZIF-8 promoting cell viability in electrospun tricuspid valve scaffolding. In Preparation.

Barrett TJ, Minus ML. Nose-Hoover integrators at-a-glance: Barostat integration has a demonstrable effect on uniaxial tension results. J. Chem. Theory Comput., 2025.

PATENT: Barrett TJ, Moldovean-Cioroianu NS, Altintas Z. Designing imprinted polymer-based functional materials for small molecules using computational methods. Pending, Germany.

HONORS & FELLOWSHIPS

NSF PACK International Research Fellowship (Award #1829573) · Christian-Albrechts-Universität zu Kiel Oct – Dec 2023